

Node.js + Express + TypeScript Enterprise Package Guide

Runtime Dependencies

Package	Purpose	When to Use
express	Web framework	REST APIs, middleware, routing
mongoose	MongoDB ODM	Database models, validation, queries
bcryptjs	Password hashing	Store passwords securely
jsonwebtoken	JWT Authentication	Access Token & Refresh Token
dotenv	Load <code>.env</code> variables	Environment configuration
envalid	Validate environment variables	Ensure required env variables exist at startup
cors	Cross-Origin Resource Sharing	Allow frontend to access backend
helmet	Security headers	Protect against common HTTP attacks
hpp	HTTP Parameter Pollution protection	Prevent duplicate query parameter attacks
xss-clean	XSS protection	Sanitize malicious user input
compression	Gzip compression	Faster API responses
cookie-parser	Parse cookies	JWT in cookies, sessions
express-rate-limit	Rate limiting	Prevent brute-force attacks
express-async-errors	Async error handling	Throw errors without try/catch everywhere
ioredis	Redis client	Cache, sessions, queues
pino	Logger	Application logging
pino-http	HTTP request logger	Log API requests/responses
prom-client	Prometheus metrics	Monitoring and observability
swagger-jsdoc	Generate OpenAPI spec	Swagger documentation
swagger-ui-express	Swagger UI	Interactive API documentation
validator	String validation	Email, URL, phone validation

Package	Purpose	When to Use
slugify	Create URL slugs	Product/category slugs
uuid / nanoid	Unique IDs	Public identifiers
yamljs	Read YAML files	Swagger YAML configuration
zod	Runtime validation	Request body, query, params validation

Optional Runtime Packages

File Upload

Package	Use
busboy	Large file streaming
multer	Small/normal uploads
sharp	Image optimization

CSV

Package	Use
csv-parser	Read CSV files
fast-csv	Read & write CSV
csv-writer	Generate CSV reports

Excel

Package	Use
exceljs	Read/write Excel (.xlsx)

Queue

Package	Use
bullmq	Background jobs
ioredis	Queue backend

Examples:

- Email sending
- Notifications
- Payroll generation
- Report generation
- CSV import

Email

Package	Use
nodemailer	SMTP email sending

Cloud Storage

Package	Use
@aws-sdk/client-s3	Amazon S3
@aws-sdk/lib-storage	Multipart uploads

Scheduling

Package	Use
node-cron	Scheduled jobs

Examples:

- Daily attendance
- Monthly payroll
- Cleanup jobs

Development Dependencies

Package	Purpose
typescript	TypeScript compiler
tsx	Run TS directly in development
tsc-alias	Resolve TypeScript path aliases after build
rimraf	Remove build folder
cross-env	Cross-platform environment variables
eslint	Code quality
prettier	Code formatting
eslint-config-prettier	Prevent ESLint & Prettier conflicts
@eslint/js	ESLint base config
typescript-eslint	TypeScript ESLint support
jest	Unit testing
ts-jest	Run Jest with TypeScript
supertest	API testing
@types/node	Node.js types
@types/express	Express types
@types/cors	CORS types
@types/cookie-parser	Cookie parser types
@types/compression	Compression types
@types/jsonwebtoken	JWT types
@types/swagger-ui-express	Swagger UI types
@types/yamljs	YAMLJS types
@types/validator	Validator types
@types/jest	Jest types
@types/supertest	Supertest types

Recommended Enterprise Packages

Package	Why
express-mongo-sanitize	Prevent MongoDB injection
mongodb-memory-server	MongoDB integration testing
husky	Git hooks
lint-staged	Lint only staged files
commitlint	Commit message validation
pino-pretty	Pretty logs in development

Packages by Module

Authentication

- bcryptjs
- jsonwebtoken
- cookie-parser
- express-rate-limit

Security

- helmet
- hpp
- cors
- xss-clean
- express-mongo-sanitize

Database

- mongoose

Cache

- ioredis
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Validation

- zod
 - validator
-

Documentation

- swagger-jsdoc
 - swagger-ui-express
 - yamls
-

Monitoring

- pino
 - pino-http
 - prom-client
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File Processing

- busboy
 - fast-csv
 - csv-parser
 - exceljs
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Background Jobs

- bullmq
 - node-cron
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Typical Enterprise Stack

- **Backend:** Express + TypeScript
- **Database:** MongoDB + Mongoose
- **Cache:** Redis (ioredis)
- **Authentication:** JWT + bcryptjs
- **Validation:** Zod
- **Logging:** Pino
- **Security:** Helmet + HPP + CORS + Rate Limiting
- **Documentation:** Swagger

- **Monitoring:** Prometheus (`prom-client`)
- **Background Jobs:** BullMQ
- **Large File Uploads:** Busboy
- **CSV/Excel:** Fast-CSV + ExcelJS
- **Testing:** Jest + Supertest